NIHAL SIDDIQUI

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SUMMARY

AI/ML Engineer building production-ready LLM and MLOps systems with FastAPI, LangGraph, Pinecone, and AWS. Experienced in deploying scalable RAG applications, multi-agent workflows, and end-to-end ML pipelines. Passionate about building reliable AI systems for real-world applications.

TECHNICAL SKILLS

Core: Python, SQL, FastAPI, REST APIs, Docker, Git
Generative AI & NLP: LLMs, RAG, LangChain, LangGraph, Pinecone, LangSmith, Embedding Models
Machine Learning & MLOps: XGBoost, Scikit-learn, TensorFlow, MLflow, Great Expectations, Optuna, Redis
Cloud & DevOps: AWS (EC2, ECR), GitHub Actions (CI/CD)

PROJECTS

Enterprise-Grade RAG System |https://github.com/nihalsiddiqui7/production-rag| **Python, FastAPI, LangChain, Pinecone, AWS**

• Built production RAG API (FastAPI + Pinecone) serving 10K+ documents with P95 latency under 400ms, 97% answer correctness, and 100% groundedness.
• Engineered ingestion pipeline with recursive text chunking; improved retrieval relevance from 70% to 95% through iterative prompt and chunk-size optimization.
• Integrated Redis caching, prompt injection defenses, PII anonymization (Presidio), and automated evaluation via LangSmith + OpenEvals (33+ runs tracked).

Multi-Agent Travel Planner | github.com/nihalsiddiqui7/travel-plan-agent **| Python, FastAPI, LangGraph, OpenAI, PostgreSQL, Docker**

• Built multi-agent travel system (LangGraph) with Flight, Hotel, and Itinerary agents coordinated via PostgreSQL checkpointing for persistent, resumable workflow state.
• Integrated Tavily API and custom airport resolution tool for real-time flight/hotel data, grounding itinerary generation in live search results.
• Deployed containerized FastAPI service with Jinja2 UI, health monitoring, and LangSmith tracing for production observability.

Customer Churn Prediction Platform | https://github.com/nihalsiddiqui7/churn-prediction **| XGBoost, MLflow, FastAPI, Docker, AWS EC2**

• Developed and deployed an ML-driven churn prediction pipeline achieving 93% recall through feature engineering and hyperparameter tuning (Optuna).
• Automated data validation (Great Expectations) and experiment tracking (MLflow), ensuring 100% reproducible workflows.
• Containerized inference engine with Docker and deployed on AWS EC2 via GitHub Actions CI/CD.
• Optimized classification thresholds with stakeholders, projecting $20.7M annual revenue retention and 30.8% operational cost reduction.

EDUCATION

MGM University, Institute of Information and Communication Technology
Bachelor of Technology in Data Science | Expected Graduation: 2027